



UNIVERSITI MALAYSIA TERENGGANU

CSE3433 SOFTWARE ARCHITECTURE

SEMESTER II SESSION 25/26

TITLE:

MICROSERVICE DISASTER RELIEF COORDINATION PLATFORM

INDIVIDUAL LOGBOOK

NAME: MUHAMMAD ADHA BIN MUHAMMAD HAMKA

MATRIC NO.: S76187

COURSE: BACHELOR OF COMPUTER SCIENCE (SOFTWARE ENGINEERING) WITH
HONOURS

LECTURE: DR. MOHAMAD NOR HASSAN

DATE: 18 JUN 2026

Team Information

PROJECT TITLE: MICROSERVICE DISASTER RELIEF COORDINATION PLATFORM

DEVELOPER: MUHAMMAD ADHA BIN MUHAMMAD HAMKA

PROJECT ROLE: PROJECT MANAGER

TECH STACK: Node.js, Express.js, MongoDB Atlas, HTML/CSS/JS, Bootstrap 5

GROUP: G02

TEAM MEMBERS: -

1. MUHAMMAD AIMAN MUKHLIS BIN KHAIRUM ANUAR
2. MUHAMMAD SYAKIR ZUFAYRI BIN SAARI

Weekly Progress

Week	Project Activity	Duration	Contribution (Individual)	Challenges	Action Plan
Week 4 – Week 5	Backend API & Database Architecture	15 hours	Engineered RESTful APIs for the Volunteer and Communication microservices. Isolated services on separate ports (9020, 6020) to ensure architectural independence. Initialized local MongoDB schemas.	Encountered silent Node.js runtime crashes due to encoding mismatches in the terminal, and microservice port conflicts.	Implement CORS middleware, finalize database schemas, and begin drafting the frontend user interface.
Week 6 – Week 7	UI/UX Dashboard Development	12 hours	Developed a responsive, single-page-application (SPA) style dashboard using Bootstrap 5. Engineered	CSS layout integrity failed when merging existing form logic into the new DOM structure.	Decouple CSS into a globally accessible dashboard.css file to strictly separate structural

			with a "Holy Grail" CSS layout featuring a fixed navigational sidebar and header for cross-service consistency.		styling from data logic.
Week 8– Week 13	System Integration & Cloud Deployment	10 hours	Integrated frontend client with backend microservices via asynchronous fetch() requests. Verified data pipelines using Postman. Migrated local database to a cloud-hosted MongoDB Atlas cluster.	Triaged ERR_CONNECTION_REFUSED errors caused by port configuration mismatches between the client's fetch requests and the Express listeners.	Secure the repository using a .gitignore file, initialize Git via Git Bash, and push the stable build to GitHub.

GenAI Usage Record

Date	AI Tool	Task / Prompt Description	AI Contribution	Human Verification & Edits
1 April 2026	Gemini	Debugging network errors: "404 request resource cannot be found" and "Cannot GET /"	Differentiated between frontend UI routing and backend API endpoints. Identify missing route parameters.	Verified backend uptime by testing the exact GET /api/v1/volunteers/match endpoint in Postman to ensure a 200 OK response before modifying frontend code.
15 April 2026	Gemini	UI Standardization: Refactoring raw HTML/CSS into a unified dashboard layout.	Provided a standardized CSS layout utilizing position: fixed and Flexbox to create a responsive sidebar and header architecture.	Conducted a code review of the generated CSS. Manually transplanted existing Volunteer and Communication form logic into the newly structured main content grid.
29 April 2026	Gemini	Environment Setup: Cloud database migration and version control.	Supplied the navigation flow for MongoDB Atlas and the exact terminal commands required to initialize and push a local Git repository.	Executed Git Bash commands manually. Verified the .gitignore file successfully excluded node_modules and the .env configuration file prior to committing.
13 May 2026	Gemini	"Explain how to implement JWT authentication in an Express microservice."	Provided a structural overview of token generation (jwt.sign) and a custom middleware function	Adapted the generic middleware to strictly check for specific 'Admin' vs 'Volunteer' roles before

			to verify tokens on protected routes.	granting access to database endpoints.
27 May 2026	Gemini	"Write a MongoDB aggregation pipeline to group inventory items by location and sum the quantities."	Generated the \$group and \$sum MongoDB syntax required to aggregate the raw supply data.	Tested the query in MongoDB Compass directly to ensure accuracy before hardcoding it into the Node.js backend route.
10 June 2026	Gemini	Deployment debugging: "Cross-Origin Request Blocked after deploying to Render."	Explained that the live server environment dynamically assigns ports and required updating the app.use(cors()) configuration to accept the specific live frontend domain.	Updated the backend .env files with the correct origin URLs and successfully tested cross-origin requests from the live dashboard.